



SQITK3073 - BUSINESS ANALYTIC PROGRAMMING

(Group 4)

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Group Project

Car Price Prediction System

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INTRODUCTION

The automotive industry is fast-changing with increased dependence on technology to enhance efficiency and reliability in transactions of the used car market. Every day, millions of used cars are displayed online, and it is very hard for both a buyer and a seller to identify a just and correct price for a vehicle. The value of a car depends on several factors: age, mileage, location, and condition; hence, manual evaluation is not only time-consuming but also subjective and inconsistent.

Online platforms like Mudah list many cars for sale, but most users don't have the tools to analyze trends or compare prices effectively. This often results in buyers overpaying for a vehicle and sellers undervaluing their cars. In this regard, there is an obvious requirement for data-driven solutions that automate price estimation, ensuring transparency and trust in transactions.

This report describes the machine learning-based car price prediction system that integrates web scraping, preprocessing, and predictive modeling with a user-friendly dashboard for real-time price predictions. The system considers various attributes of the car, including year, mileage, and location, to provide precise price insights for buyers and sellers. At the heart of this solution lies a web scraper, which gathers car data directly from Mudah, ensuring the predictions are based on real-world market conditions. This approach ensures transparency, efficiency, and fairness—some of the challenges associated with inconsistent pricing and complexity in data are solved.

PROBLEM STATEMENT

The process to determine the fair market value of a used car is quite convoluted and driven by the mix of subjective judgment and the inherent complexity in data factors such as mileage, year, location, and market demand. Some of the key challenges include:

1. Inconsistent Pricing:

- With different opinions, there is inconsistency in pricing; hence, buyers and sellers can't agree due to the perception of value—very difficult and frustrating for buyers and sellers.

2. Data Complexity:

- Multiple variables, including the car's condition, mileage, age, and location, are interlinked in determining price. Analyzing those manually across hundreds of listings is inefficient and mostly inaccurate.

3. Transparency Gap:

- Buyers and sellers often do not have access to historical data and market trends, which generates mistrust and unfair practices. Sellers may undervalue their cars due to insufficient data, while buyers may overpay for vehicles that appear overpriced.

4. Limited Automation:

- The current estimation tools are either too simple or not specific to regional markets, hence missing local demand and supply nuances.

These problems bring about the necessity for a holistic automated solution that can effectively forecast car prices using the most state-of-the-art techniques in machine learning with past data. The ideal system must then be user-friendly, scalable for huge datasets, and powerful enough to hold up under ever-changing market conditions.

PROPOSED SOLUTION

The proposed solution will try to overcome these challenges by developing a machine learning-based car price prediction system. It integrates the components described below:

1. Data Collection:

- Web scrapping of the Mudah website has been done using Selenium. It captures all important features, including price, mileage, year of manufacture, location, and URL.
- The designed scraper is able to move through dynamic web pages, wait for elements to load, and scroll to fetch all relevant listings.
- The extracted data is stored in a structured CSV format for preprocessing and modeling.

2. Data Cleaning and Preprocessing:

- Price values were cleaned by removing the currency symbols and commas before conversion to numeric format.
- Mileage ranges were converted to averages for numerical consistency.
- Missing year values were replaced with the dataset's median, and remaining missing data rows were dropped.

3. Machine Learning Model:

- A RandomForestRegressor was trained using features such as year, mileage, and location.
- The pipeline integrated preprocessing steps, including one-hot encoding for the categorical location variable.

4. Interactive Dashboard:

- A user-friendly interface was built using Streamlit to allow users to input car details and receive instant price predictions.
- The dashboard ensures accessibility and real-time usability for diverse users.

METHODOLOGY & WORKFLOW

Understanding the methodology or workflow is essential to grasp how data from the Mudah.my website was extracted, cleaned, and trained to develop a model capable of providing accurate price predictions. In our project, we conducted 4 major steps to ensure the output result of the code is optimized and provides meaningful insights to users. Moreover, these steps establish a consistent method that improves the performance of the model, ensuring that the predictions are reliable and actionable by the users.

1. Web Scraping

In generic terms, web scraping functions to extract a large amount of data from any websites that are open-sourced. Web scraping automatically organizes unstructured data, such as data in HTML content, into a structured one by extracting and changing it into a proper format such as converting it into a CSV file. This means that one does not need to copy the data manually or goes through some lengthy process. Hence, the collection of data is much more efficient and organized.

One of the most famous scrapping tools is Selenium, where we choose to use this to scrap details from Mudah.my like car's title, price, location, year, mileage as well as URL.

During the coding session, the script uses Selenium WebDriver with a Chrome browser with options to bypass automation detection and handle interruptions effectively. It then fetches details of car listing, such as title, price, location, year, mileage, and URL link and save it to a CSV file. Thus, we can see that the script is built to handle websites nicely with dynamic elements, errors, and interruptions. The process will then continue with a random delay and retrieval and is repeated until there are no more results or the user chooses to stop it.

2. Data Cleaning

The second most crucial step is data cleaning. Data cleaning is essential to find and rectify records that are found to be inaccurate, incomplete or irrelevant. One of the prominent cleaning methods is removing unwanted characters, maintaining consistent formatting, transforming data into standardized structures, and managing invalid and missing values. Also, the preparation of the numerical field involves normalization for analysis and removal of rows with incomplete information. This cleaning process attempts to convert the data into a structured and reliable format that can be used for the creation and training of machine learning models. In fact, this important step helps improve the quality and predictive power of the final model.

In the script, price column is corrected by removing the "RM" prefix and commas, converting the value into numerical format, and treating any missing value within the price column. Besides, the mileage column includes ranges such as "20,000 - 30,000", and a helper function calculates the average of the range. It then splits the range into lower and upper bounds, converts them into integers, and calculates the mean. Furthermore, missing values extracted from the year column are imputed using the median value. This process ensures that the data used for prediction is accurate and valid.

3. Model Training

Model Training is the stage where the cleaned datasets are supplied to the system where the learning takes place. During this process, the machine learning model that is developed starts to identify patterns and relationships within the data by optimizing its parameters via the input and output variables. It consists of data division into training-testing subsets, which means a model can learn from one segment and give results on an unknown one. This will allow this model to work correctly in a real-world scenario during training.

When compared to the car price prediction system, the machine learning model learns to estimate car prices based on specific features mentioned above. The process begins with the preparation of data, where information is cleaned and structured so that the features are in a suitable format for training. Numerical encoding has been used to encode categorical variables such as location. Afterwards, the dataset was divided into two subsets which are a training set and a validation set. This would teach the model how the input features relate to automobile pricing based on this data and testing data would be utilized in order to validate the model for generalization abilities. Looking at these, this involves using a Random Forest Regressor without overfitting. The model learns patterns in training data by adjusting its parameters to reduce prediction errors and combining findings from numerous decision trees to improve accuracy. In the end, a trained model is generated and run whereby one can use to predict the price as wished to.

4. Model Deployment

The last step of the methods is to deploy the trained model as an interactive web or in other words, creating dashboard to allow users to input car details and get price predictions. Hence, model deployment has always been a very important step as it functions to put machine learning models into production.

One of the most used web frameworks is Streamlit, which provides user-friendly interactive for users to interact with the model. Streamlit is also an open-source Python library that lets data scientists and machine learning engineers quickly build beautiful, custom web applications with minimal code. Its main use cases include data visualization and exploration of datasets by deploying dashboards and user interfaces with less knowledge in web development. In this project, Streamlit helps to process input from the user. For example, the year of the car, mileage, and location-and returns real-time forecasts of automobile prices. This definitely helps to visualize the data and result in an interactive way and easier for any user to understand it.

RESULTS & ANALYSIS

There are three python code files within the folder. It is important to note that each Python file operates with its own function. As stated in the methodology and workflow section, the Python file named app.py is specifically used to run the Streamlit web framework to display the dashboard. Next, cleaning_data.py is used to clean the data and provide better accuracy data to be run by app.py. Mudah_scraper.py on the other hand is used to scrape data from Mudah.my, including the year of manufacture, mileage, price and location. Together, these files collaborate to create an end-to-end solution that collects, processes, and visualizes car price predictions seamlessly.

Example Output:

```
PS C:\Users\angzh\Desktop\group assignment selenium> python mudah_scraper.py
Data will be saved to: mudah_cars.csv

Scraping page 1: https://www.mudah.my/malaysia/cars-for-sale?o=1&q=perodua
Scrolling to bottom of page...
Extracted: Perodua ALZA 1.5 SXi (M) SUNROOF *SEMI BUCKET SEAT - RM 20,800
Extracted: 2019 Perodua MYVI 1.5 AV ADVANCE (A) - RM 39,800
Extracted: 2018 Perodua MYVI 1.3 PREMIUM X (A) - RM 35,875
Extracted: 2019 Perodua MYVI 1.3 G (A) - RM 35,800
Extracted: 2016 Perodua BEZZA 1.3 PREMIUM X (A) - RM 26,300
Extracted: Perodua BEZZA 1.0 GXTRA (A) 2020 **ON THE ROND - RM 27,800
Extracted: 2019 Perodua AXIA 1.0 GXTRA (A) - RM 13,000
Extracted: 2018 Perodua MYVI 1.5 H (A) - RM 33,000
Extracted: 2021 Perodua MYVI 1.5 H (A) - RM 45,000
Extracted: 2015 Perodua MYVI 1.5 SE FACELIFT (A) - RM 626/month
Extracted: 2014 Perodua MYVI 1.3 SE ZHS (A) - RM 545/month
Extracted: 2005 Perodua KELISA 1.0 EZ (A) - RM 15,500
Extracted: 2011 Perodua MYVI 1.5 EXTREME ZHX (A) - RM 16,723
Extracted: 2014 Perodua VIVA 1.0 (A) - RM 11,000
Extracted: 2020 Perodua MYVI 1.5 ADVANCE (A) FULL SERVICE - RM 41,800
Extracted: 2021 Perodua AXIA 1.0 AV (A) - RM 33,000
Extracted: 2013 Perodua MYVI 1.5 SE ZHS (A) - RM 22,500
Extracted: 2020 Perodua BEZZA 1.3 X PREMIUM (A) - RM 34,888
Extracted: 2005 Perodua MYVI 1.3 SX (M) - RM 7,500
Extracted: 2013 Perodua VIVA 1.0 ELITE EZi (A) - RM 14,000
Extracted: Perodua BEZZA 1.3 PREMIUM X D63D GearUp HIGHSPEC - RM 28,800
Extracted: 2007 Perodua MYVI 1.3 EZi (A) - RM 10,900
Extracted: 2023 Perodua ATIVA 1.0 H (A) LIKE NEW CONDITION - RM 53,800
Extracted: 2015 Perodua MYVI 1.5 SE FACELIFT (A) 🟢ICON - RM 28,800
Extracted: 2015 Perodua ALZA 1.5 ZV ADVANCED FL-2 YRS WARRANTY - RM 30,888
Extracted: 2016 Perodua AXIA 1.0 G SPEC (A) PREMIUM - RM 22,888
Extracted: 2021 Perodua MYVI 1.3 PREMIUM X PRE-CRASH-WARRANTY - RM 37,888
Extracted: Perodua AXIA 1.0 G (A) 2021 **ON THE ROND - RM 26,800
Extracted: 2011 Perodua MYVI 1.3 SE FACELIFT (A) - RM 28,800
Extracted: Perodua Myvi 1.3 (A) High Spec Sporty Hatchback - RM 33,800
Extracted: 2017 Perodua BEZZA 1.3 ADVANCE (A) - RM 27,500
Extracted: Perodua Myvi 1.5 SE (A) Facelift High Special Spec - RM 30,800
Extracted: 2022 Perodua MYVI 1.5 AV (A) PROMO OTR 4K MILE SHJ - RM 49,988
Extracted: 2020 Perodua BEZZA AV 1.3 (A) PROMO OTR MILE 14K - RM 50 988
Extracted: 2021 Perodua BEZZA 1.0 (A) PROMO OTR 12k MILES SHJ - RM 39 988
```

Figure 1: Data Scraping Process

```

Scraping page 2: https://www.mudah.my/malaysia/cars-for-sale?o=2&q=perodua
Scrolling to bottom of page...
Extracted: 2008 Perodua MYVI 1.3 EZi (A) AIRBAG LOW MILEAGE - RM 13,300
Extracted: 2011 Perodua MYVI 1.3 EZi (A) AIRBAG TIP TOP - RM 14,800
Extracted: 2021 Perodua ALZA 1.5 SE (A) 1K MUKA TIP TOP - RM 48,800
Extracted: 2010 Perodua MYVI 1.3 EZL (LIMITED EDITION) (A) - RM 12,500
Extracted: 2023 Perodua ALZA 1.5 AV MY22 (A) - RM 70,800
Extracted: 2020 Perodua AXIA 1.0 SE (A) - RM 29,800
Extracted: 2011 Perodua MYVI 1.3 SE ZHS FACELIFT (A) - RM 15,900
Extracted: 2002 Perodua KELISA 1.0 EZ (A) - RM 7,900
Extracted: kenari auto 2002 - RM 7,999
Extracted: 2020 Perodua ARUZ 1.5 ADVANCE (A) LEATHER SEAT - RM 48,800
Extracted: 2021 Perodua BEZZA 1.3 X (A) FACELIFT PUSH START - RM 35,800
Extracted: 2013 Perodua VIVA 1.0 ELITE EZi (A) - RM 9,990
Extracted: 2016 Perodua AXIA 1.0 E (M) - RM 13,888
Extracted: 2010 Perodua VIVA 1.0 EZi (A) - RM 9 888
Extracted: 2011 Perodua ALZA 1.5 EZ (A) - RM 22,500
Extracted: 2011 Perodua MYVI 1.3 EZi (A) - RM 1,764/month
Extracted: 2018 Perodua ALZA 1.5 ZV AV FACELIFT (A) BODY KIT - RM 39,800
Extracted: 2018 Perodua MYVI 1.3 PREMIUM X (A) - RM 422/month
Extracted: 2013 Perodua MYVI 1.5 EXTREME ZHX (A) - RM 15,700
Extracted: 2018 Perodua MYVI 1.5 ADVANCE (A) F/S PERODUA - RM 41,900
Extracted: Perodua MYVI 1.5 AV (A) FULLSERVIS PERODUA WARRANTY - RM 49,999
Extracted: 2011 OTR MYVI 1.3 SE (A) Mileage 113K F/S PERODUA - RM 19 800
Extracted: 2006 Perodua KELISA 1.0 EZ (A) - RM 15,000
Extracted: 2020 Perodua MYVI 1.3 STANDARD G (A) - RM 34,800
Extracted: Perodua AXIA 1.0 E (M) - RM 18,500
Extracted: 2018 Perodua MYVI 1.3 PREMIUM X 3 Year Warranty - RM 31,800
Extracted: 2021 Perodua MYVI 1.5 AV 14k Mileage/Perodua Serv - RM 46,800
Extracted: 2010 Perodua VIVA 1.0 ELITE EXCLUSIVE (A) - RM 8,300
Extracted: 2013 Perodua MYVI 1.5 SE ZHS (A) - RM 19,500
Extracted: 2010 Perodua ALZA 1.5 EZi (A) - RM 21,900
Extracted: 2010 Perodua ALZA 1.5 EZi (A) - RM 12,000
Extracted: 2014 Perodua ALZA SE 1.5 A Facelift/F.LOAN/Bodykit - RM 32,999
Extracted: 2012 Perodua MYVI 1.3A EZi PREMIUM - RM 19,800
Extracted: 2014 Perodua MYVI 1.3 SXi PREMIUM (M) - RM 12,800
Extracted: 2025 Perodua AXIA 1.0 G CVT - PROMOTION - RM 38,380
Extracted: Y 2016 Perodua ALZA 1.5 ADVANCED FACELIFT (A) - RM 32,800
Extracted: Confirm FullLoan New Perodua AXIA ReadyStock - RM 35,999
Extracted: 2017-Perodua BEZZA 1.3 ADVANCE (A)Kedai/Bank/Cash - RM 31,990
Extracted: 2015 Perodua MYVI 1.5 SE FACELIFT (A) - RM 31,800
Extracted: Perodua MYVI 1.3 SE (A) 2008 OREN LAMBO - RM 18,900
Wrote 40 cars to CSV
Total cars scraped so far: 80

```

Figure 2: The Continuation of Data Scraping

Explanation:

The figures above show part of the scrapping process by Selenium. We can denote that when the code file (Mudah_scraper.py) is run, it is scrapping and extracting all the information from Mudah.my. It was then recorded all the details in a CSV file and the total car scrap within the dedicated page is written. For instance, on the first page, the cars scraped is 80 cars.

```

Extracted: 2025 Perodua ATIVA AV MY21 D55L (A) - RM 72,600
Extracted: 2021 Perodua MYVI 1.5 AV (A)FSR/PERAK/TIPTOP - RM 46,300
Extracted: 2013 Perodua MYVI 1.5 EXTREME ZHX (A). Full Loan - RM 25,888
Wrote 40 cars to CSV
Total cars scraped so far: 3480

Scraping page 88: https://www.mudah.my/malaysia/cars-for-sale?o=88&q=perodua
Scrolling to bottom of page...
Extracted: 2018 Perodua MYVI 1.5 ADVANCE (A)Loan senang lulus - RM 42,900
Extracted: 2019 Perodua MYVI 1.5 (A) AV 1 Year Warranty - RM 46 800
Extracted: 2018 Perodua Myvi 1.5 Advance 5-YRS Warranty - RM 42,800
Extracted: 2020 Perodua BEZZA 1.0 GX (M)LOAN KEDAI - RM 32,800
Extracted: 2019 Perodua AXIA 1.0 ADVANCE AV (A) LOW DEPO - RM 32,800
Extracted: 2021 Perodua Myvi 1.5 F/Service 59k-Mileage Only - RM 45,800
Extracted: 2021 Perodua Axia 1.0 GX (A) 5-YRS Warranty - RM 29,800
Extracted: 2016 Perodua AXIA 1.0 (A) TIPTOP CONDITION - RM 20,800
Extracted: READY STOCK 2024 Perodua AXIA 1.0 X - RM 39,780
Extracted: AXIA 1.0 G auto READY STOCK - RM 38,380
Extracted: 2013 Perodua VIVA 1.0 ELITE AT (A) - RM 11,800
Extracted: 2017 Perodua BEZZA 1.3 PREMIUM X (A) - RM 34,800
Extracted: New Perodua Axia Fast Stock & Mudah Lulus - RM 38,600
Extracted: 2019 Perodua Bezza 1.3 X (A) 5-Years Warranty - RM 33,800
Extracted: Perodua MYVI 1.3 PREMIUM X (A) 47k km Loan Kedai - RM 33,700
Extracted: Perodua Aruz Ready Stock & Mudah Lulus - RM 77,600
Extracted: Perodua ATIVA AV (A) 2025 STOCK TERSEDIA FULLOn - RM 70,000
Extracted: 2014 Perodua MYVI 1.3 EZi PREMIUM (A)FULLOAN 0DEPO - RM 23,900
Extracted: 2014 Perodua MYVI 1.3 EZi PREMIUM (A)FULLOAN - RM 25,800
Extracted: 2009 Perodua MYVI 1.3 FACELIFT (A)•BOLEH LOAN - RM 14,688
Extracted: 2006 Perodua MYVI 1.3 EZi (A) - RM 14,900
Extracted: 2005 Perodua KELISA 1.0 EZ (A) - RM 9,999
Extracted: 2022 Perodua BEZZA 1.3 AV (A) LOW MIL 12K - RM 40,888
Extracted: 2015 Perodua ALZA 1.5 EZ FACELIFT (A) 1 LADY OWNER - RM 27,888
Extracted: 2016 Perodua ALZA 1.5 EZ FACELIFT (A) F/WRNTY - RM 28,888
Extracted: 2018 Perodua BEZZA 1.3 ADVANCE (A) REVERSE CAMERA - RM 31,888
Extracted: 2017 Perodua BEZZA 1.3 ADVANCE (A) ONE OWNER - RM 33 888
Extracted: 2013 Perodua ALZA 1.5 EZ (A) ONE OWNER - RM 21,888

```

Figure 3: End of Scraping Process

Explanation:

The figure above shows the end of scrapping process by the user. From the terminal output, we can see that the total pages and cars scraped are summarized at the bottom of the code. Besides, it also mentioned that all the data are saved into a CSV file named mudah_cars.csv.

```
app.py activate.bat car_price_predictor.py clearing_data.py mudah_scraper.py mudah_cars.csv X base.py
mudah_cars.csv
29 2022 Perodua ATIVA 1.0 H (A), "RM 53,000", Selangor, 2022, 5000 - 9999, https://www.mudah.my/2022-perodua-ativa-1-0-h-a-109764347.htm
30 2014 Perodua ALZA 1.5 ZHV ADVANCED (A), "RM 24,800", Melaka, N/A, 180000 - 189999, https://www.mudah.my/2014-perodua-alza-1-5-zhv-advanced-a-109764323.htm
31 2008 Perodua KENARI 1.0 EZ AERO (A), "RM 12,000", Melaka, 2008, 45000 - 49999, https://www.mudah.my/2008-perodua-kenari-1-0-ez-aero-a-109764311.htm
32 2014 Perodua ALZA 1.5 AT EZI/EZ/SE/AV KEADI/ADJUST, "RM 19,800", Selangor, N/A, 130000 - 139999, https://www.mudah.my/2014-perodua-alza-1-5-at-ezi-ez-se-av-keadi-adjust-109567643.htm
33 2015 Perodua MYVI 1.5 SE FACELIFT (A), "RM 37,800", Sabah, 2015, 100000 - 109999, https://www.mudah.my/2015-perodua-myvi-1-5-se-facelift-a-109840201.htm
34 Perodua VIVA 1.0 ELITE (Auto) 2010, "RM 9,300", Negeri Sembilan, 2010, 90000 - 94999, https://www.mudah.my/perodua-viva-1-0-elite-auto-2010-109764110.htm
35 2013 Perodua MYVI 1.3 SE ZHS (A) SIAP WARRANTY!!!!, "RM 19,000", Melaka, 2013, 110000 - 119999, https://www.mudah.my/2013-perodua-myvi-1-3-se-zhs-a-siap-warranty-109557713.htm
36 2020 Perodua MYVI 1.5 H FULL SERVICE WARRANTY, "RM 42,999", Selangor, 2020, 80000 - 84999, https://www.mudah.my/2020-perodua-myvi-1-5-h-full-service-warranty-109635519.htm
37 2006 Perodua KANCIL 659cc 660 BX BASIC ENHANCED, "RM 2,500", Selangor, N/A, 90000 - 94999, https://www.mudah.my/2006-perodua-kancil-659cc-660-bx-basic-enhanced-109764066.htm
38 2019 Perodua MYVI 1.5 AV (A) DEPO 300 SAJA, "RM 44,888", Kuala Lumpur, 2019, 85000 - 89999, https://www.mudah.my/2019-perodua-myvi-1-5-av-a-depo-300-saja-109763924.htm
39 2016 Perodua MYVI 1.3 PREMIUM X (A) loan kedai 1000, "RM 23,900", Selangor, 2016, 75000 - 79999, https://www.mudah.my/2016-perodua-myvi-1-3-premium-x-a-loan-kedai-1000-109763931.htm
40 2018 Perodua MYVI 1.5 ADVANCE (A), "RM 541/month", Johor, 2018, 50000 - 54999, https://www.mudah.my/2018-perodua-myvi-1-5-advance-a-108328862.htm
41 2018 Perodua AXIA 1.0 G FACELIFT (A), "RM 281/month", Selangor, 2018, 65000 - 69999, https://www.mudah.my/2018-perodua-axia-1-0-g-facelift-a-109701363.htm
42 2011 Perodua MYVI 1.3 EZI FACELIFT (A), "RM 1,176/month", Selangor, 2011, 95000 - 99999, https://www.mudah.my/2011-perodua-myvi-1-3-ezi-facelift-a-108928982.htm
43 2015 Perodua AXIA 1.0 AV (A) FULL SPEC-1 OWNER, "RM 23,800", Selangor, 2015, 90000 - 94999, https://www.mudah.my/2015-perodua-axia-1-0-av-a-full-spec-1-owner-106448750.htm
44 2006 Perodua KANCIL 847cc 850 EX ENHANCED (H), "RM 4,000", Terengganu, 2006, 300000 - 349999, https://www.mudah.my/2006-perodua-kancil-847cc-850-ex-enhanced-m-109764040.htm
45 2017 Perodua MYVI 1.5 SE FACELIFT (A), "RM 32,000", Sarawak, 2017, 130000 - 139999, https://www.mudah.my/2017-perodua-myvi-1-5-se-facelift-a-109764039.htm
46 2012 Perodua ALZA 1.5 EZI (A), "RM 25,000", Johor, 2012, 15000 - 19999, https://www.mudah.my/2012-perodua-alza-1-5-ezi-a-109764020.htm
47 2014 Perodua ALZA 1.5 SE ZS FACELIFT (A), "RM 30,000", Sarawak, 2014, 130000 - 139999, https://www.mudah.my/2014-perodua-alza-1-5-se-zs-facelift-a-109764015.htm
48 Perodua 2025 MYVI 1.5 AV, "RM 59,310", Selangor, N/A, 0 - 4999, https://www.mudah.my/perodua-2025-myvi-1-5-av-93434730.htm
49 2019 Perodua AXIA 1.0 STANDARD G (A), "RM 26,800", Johor, N/A, 65000 - 69999, https://www.mudah.my/2019-perodua-axia-1-0-standard-g-a-107308121.htm
50 Perodua ALZA 1.5 SE FACELIFT Android R/Cam LikeNEW, "RM 36,800", Selangor, 2017, 65000 - 69999, https://www.mudah.my/perodua-alza-1-5-se-facelift-android-r-cam-likeneu-109067270.htm
51 2025 Perodua BEZZA 1.3 X MY16 FACELIFT (A), "RM 45,980", Johor, N/A, 0 - 4999, https://www.mudah.my/2025-perodua-bezza-1-3-x-my16-facelift-a-101481094.htm
52 2025 new Perodua ALZA 1.5 X Fast Stock, "RM 64,933", Johor, N/A, 0 - 4999, https://www.mudah.my/2025-new-perodua-alza-1-5-x-fast-stock-104164498.htm

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Extracted: 2021 Perodua BEZZA 1.3 X (A) FACELIFT PUSH START - RM 35,800
Wrote 40 cars to CSV
Total cars scraped so far: 160

Scraping page 5: https://www.mudah.my/malaysia/cars-for-sale?o=5&q=perodua
Scrolling to bottom of page...
Extracted: 2013 Perodua VIVA 1.0 ELITE EZI (A) - RM 9,990
Extracted: 2016 Perodua AXIA 1.0 E (M) - RM 13,888
Extracted: 2010 Perodua VIVA 1.0 EZI (A) - RM 9,888
Extracted: 2011 Perodua MYVI 1.5 EZ (A) - RM 22,500
Extracted: 2011 Perodua MYVI 1.3 EZI (A) - RM 1,764/month
Extracted: 2018 Perodua ALZA 1.5 ZV AV FACELIFT (A) BODY KIT - RM 39,800
Extracted: 2018 Perodua MYVI 1.3 PREMIUM X (A) - RM 422/month
Extracted: 2013 Perodua MYVI 1.5 EXTREME 20X (A) - RM 15,700
Extracted: 2018 Perodua MYVI 1.5 ADVANCE (A) F/S PERODUA - RM 41,000
Extracted: Perodua MYVI 1.5 AV (A) FULLSERVICE PERODUA WARRANTY - RM 49,999
Extracted: 2011 QTR MYVI 1.3 SE (A) Mileage 113K F/S PERODUA - RM 19,800
Extracted: 2006 Perodua KELISA 1.0 EZ (A) - RM 15,000
Extracted: 2020 Perodua MYVI 1.3 STANDARD G (A) - RM 34,800
Extracted: Perodua AXIA 1.0 E (M) - RM 18,500
Extracted: 2018 Perodua MYVI 1.3 PREMIUM X Year Warranty - RM 31,800
Extracted: 2021 Perodua MYVI 1.5 AV Mileage/Perodua Serv - RM 46,800
Extracted: 2010 Perodua VIVA 1.0 ELITE EXCLUSIVE (A) - RM 8,300
Extracted: 2013 Perodua MYVI 1.5 SE ZHS (A) - RM 19,500
Extracted: 2010 Perodua ALZA 1.5 EZI (A) - RM 21,800
Extracted: 2010 Perodua ALZA 1.5 EZI (A) - RM 12,000
```

Figure 4: Output of CSV files

Explanation:

The figure above shows the terminal output in the CSV file named mudah_cars.csv. From the output, we can definitely be sure that the data scrapped were recorded in a CSV file and contains all the details that are scrapped. It is also noted that the CSV file generates itself when the scraping process is being done.

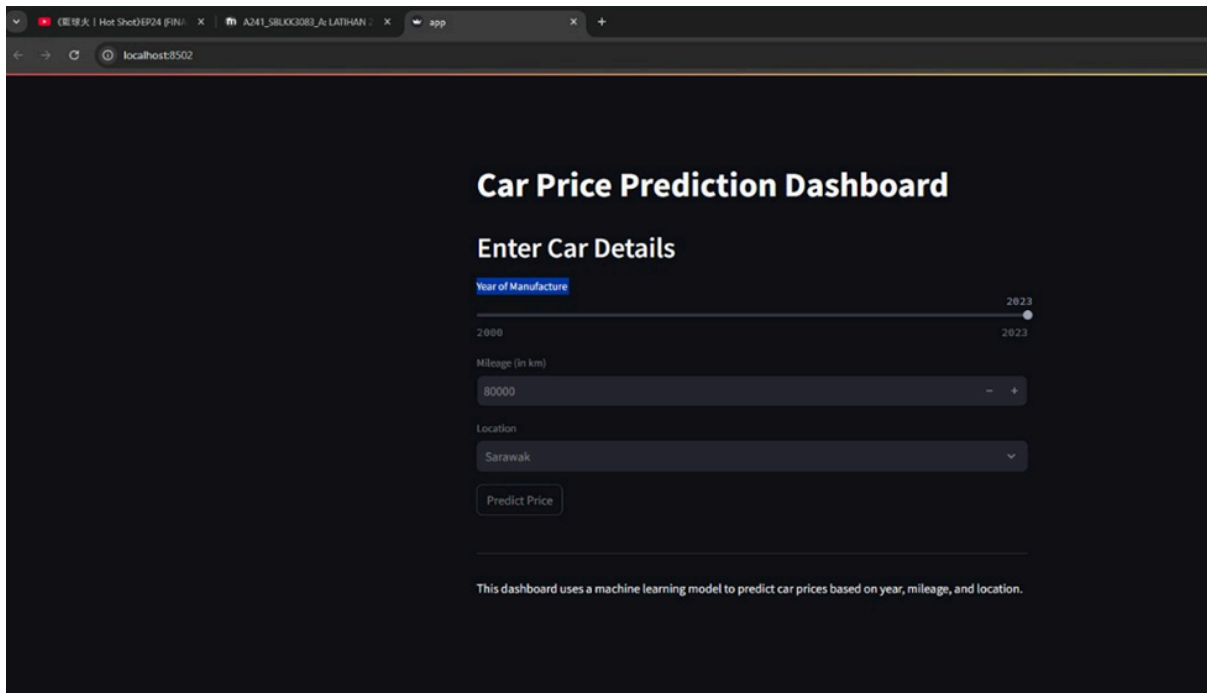


Figure 5: Dashboard Example

Explanation:

The figure above shows the interactive dashboard when the Python file containing Streamlit web framework is run. From the output, we can see that all the coded details are present and the user is able to input years of manufacture, mileage, or location to predict the car price by using machine learning model.

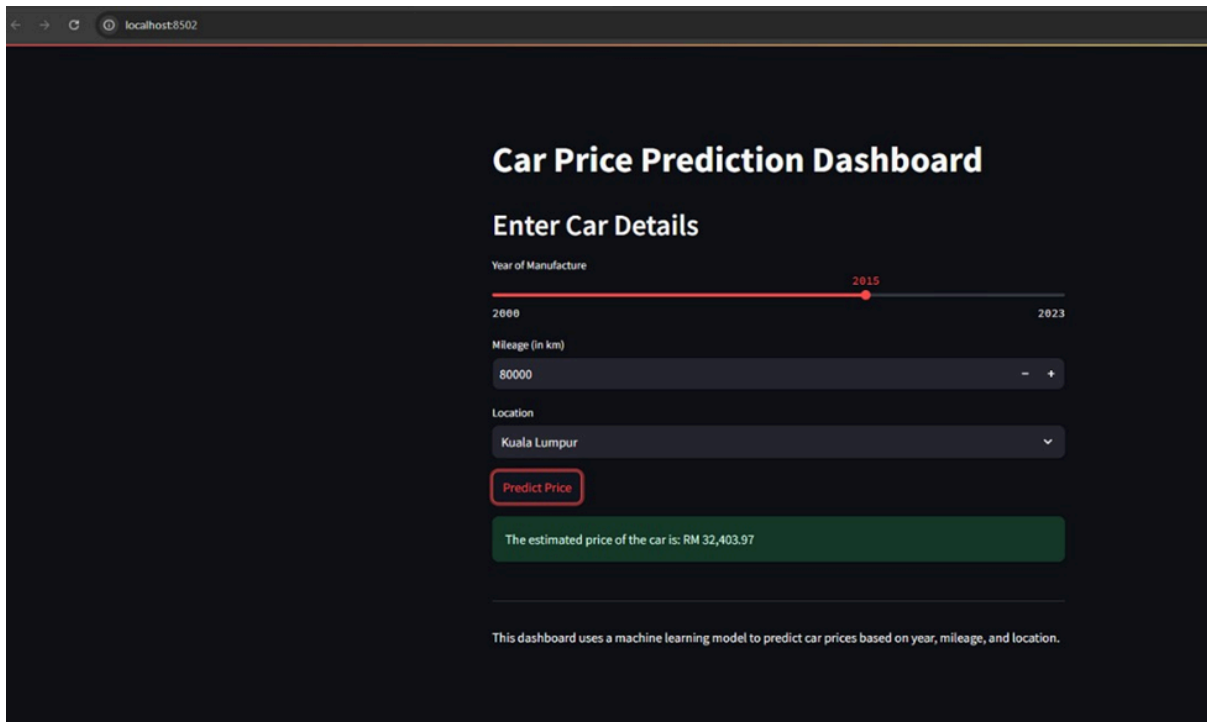


Figure 6: Example Output After Interacting With the Dashboard

Explanation:

Figure above shows the output when the user clicks on the “Predict Price” button after inputting all the necessary details. The estimated price of the car is shown, highlighted in green.

CONCLUSION

The car price prediction system has been able to address all the problems stated above, especially its complexities and inconsistencies of pricing in the used car market. It uses online scraping such as Selenium, data preprocessing, and machine learning techniques to come up with an automated and reliable method of calculating car pricing. This eliminates subjective judgment and makes the process transparent, hence building confidence between buyers and sellers.

Besides, the implementation of Random Forest Regressor model definitely takes in all important factors such as year, mileage and location. The strength of the Random Forest Regressor is to handle complex, nonlinear correlations between these features and the target variable, which is the vehicle price, thereby preventing overfitting and providing accurate and reliable predictions. Therefore, the regression model helped to analyze the relationship between all the factors and their impact on the price of the car.

Next, the use of Streamlit dashboard also enhances user accessibility and friendliness interactions which increases the reliability and real-time interactions towards the dashboard. Besides, interactive design simplifies the presentation of data, hence it enables users to conceptualize better the expected outcomes and trends. This combination of capability and accessibility enhances the stability of the system, further ensuring a seamless and engaging user experience.

In conclusion, the system would have the potential to simplify the overview pricing of the car for useful information to be provided to users in buying and selling within the environment. It also has the potential to modernize the existing market by incorporating future improvements such as expanding data sources and upgrading the current model into better versions. These enhancements could improve accuracy, respond to changing market patterns, and eventually transform the way the automobile industry is evaluated.